

Qatar-1b

R-0602

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_231008 · created 2026-07-25T04:08:34+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2460225.7883 +/- 0.0027 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.189 +/- 0.056
Transit depth (Rp/Rs)^2	3.57 +/- 2.12 [%]
Orbital Inclination (inc)	85.8 +/- 2.89
Airmass coefficient 1 (a1)	0.0088 +/- 0.0079
Airmass coefficient 2 (a2)	3.0 +/- 0.97
Scatter in the residuals of the lightcurve fit is	230.23 %
Best Comparison Star	None
Optimal Aperture	0.76
Optimal Annulus	3.8
Transit Duration (day)	0.075 +/- 0.015

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.189 ± 0.056 vs 0.1463 (deviation 0.76σ)
Duration fit vs published	0.075 d vs 0.0692 d (deviation 0.39σ)
Mid-time O–C	-5.1 min
Depth SNR	3.4
Residual scatter	230.23 %

Light curve

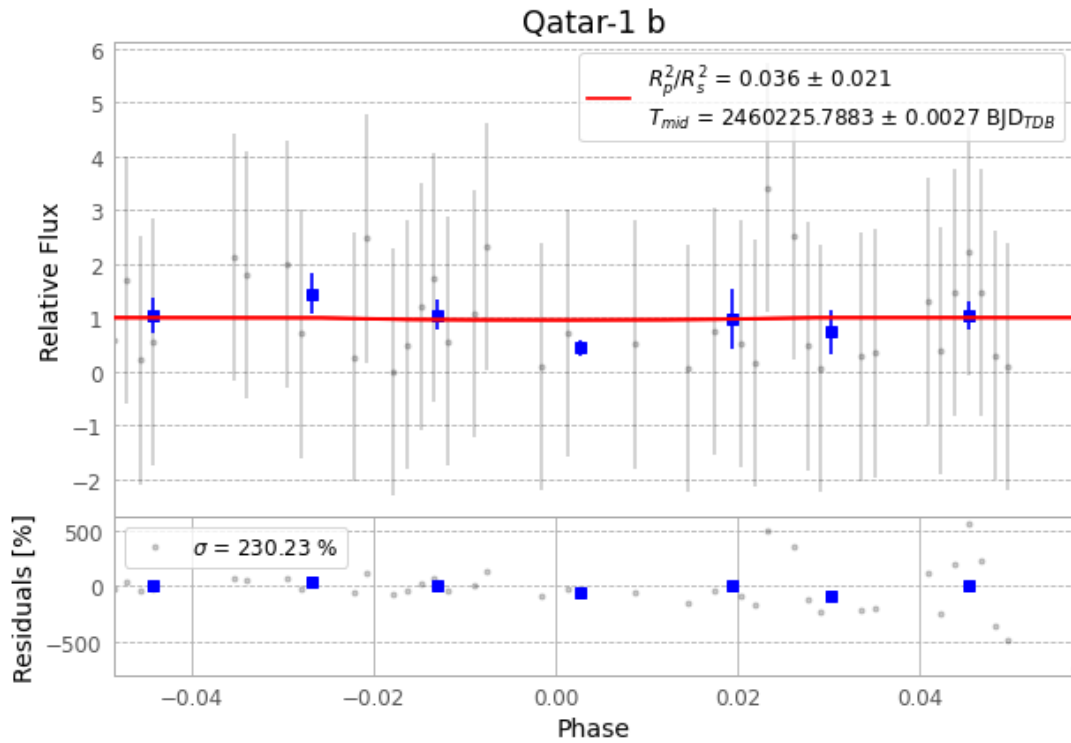


Figure 1 — FinalLightCurve_Qatar-1 b_2023-10-07.png (SHA-256 8ea7c34a4aad7595...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [460, 149], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 6ad727e30ba43242...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit b365494

Data provenance

73 FITS frames (48 MB), Qatar-1231008051215.FITS → Qatar-1231008084815.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-231008.log (9d7d1c878ed8...), FinalLightCurve_Qatar-1 b_2023-10-07.png (8ea7c34a4aad...), FinalLightCurve_Qatar-1 b_2023-10-07.csv (5859540bbf8e...)

Compute resources

Wall time 200.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-25 04:08Z.